

Homelab Blueprint Guide

Deploying VMware ESXi 8.0 and Instantiating Ubuntu Linux Virtual Machines

This technical blueprint serves as a definitive resource documenting the end-to-end orchestration of your local environment. It details the access mechanics of the VMware ESXi Direct Console User Interface (DCUI), host network mapping, storage allocation, and tactical interaction with the text-based Ubuntu Server deployment subsystem.

Phase 1: Accessing the ESXi Host Client

Following a successful host boot cycle, the physical node (Dell OptiPlex) launches the local graphical environment known as the Direct Console User Interface (DCUI), presenting system status alongside its assigned local management IP address.

CRITICAL ARCHITECTURAL PRINCIPLE

Do not manually adjust your access workstation (laptop) network configuration interface to mirror the precise address of the ESXi hypervisor. Assigning matching addresses causes immediate network contention (IP clashing). Keep the client laptop configured for **Automatic (DHCP)** network tracking.

1. Ensure your laptop and the physical ESXi server are bridged across the same local area network domain.
2. Launch a standard modern web browser and navigate directly to the target URL: `https://192.168.50.240`.
3. Bypass the browser's cryptographic boundary check (the *"Your connection is not private"* exception alert) by selecting **Advanced** and opting to **Proceed to 192.168.50.240 (unsafe)**. This allows connection using the self-signed certificate natively produced by the hypervisor host.
4. Authenticate at the blue VMware login control prompt using the primary administrative account credential identifier: `root`.

Phase 2: Hypervisor Storage & ISO Management

Virtual instances require structural assembly media during installation. An **ISO Image File** is an exact, sector-by-sector software replication of an entire physical optical disc architecture containing crucial boot metadata block sequences.

- **Execution Path:** From the main navigation index, proceed to **Storage** → select the active volume labeled **datastore1** → select the **Datastore Browser** engine located on the action banner.
- **Upload Action:** Initialize the system file directory system upload stream. Target the local repository staging path containing your production image archive files (e.g., `E:\Dummy mware`) and choose the designated installer instance file: `ubuntu-26.04-live-server-amd64`.

- **Internal State Transition:** This stream stages the image binary map directly from the workspace terminal over your network switches into the OptiPlex's physical local volume arrays, allowing instantaneous localized reads.

Phase 3: Provisioning the Virtual Machine Shell

Before launching an OS installer, a resource framework must be defined inside the system console mapping physical hardware blocks to virtual controllers.

1. Navigate to the top-level directory root or the **Virtual Machines** submenu index, and execute the **Create/Register VM** action.
2. Select the primary operational choice to **Create a new virtual machine** and advance.
3. Define your virtualization parameters:
 - **Identification Name:** Set to a standardized format string like `Ubuntu-server-test`.
 - **Guest OS Family Type:** Set value to **Linux**.
 - **Guest OS Version Matrix:** Filter and lock choice to **Ubuntu Linux (64-bit)**.
 - **Resource Boundaries:** Compute and map **2 CPUs, 4 GB RAM** (verify metrics unit flag scales to **GB**, not MB), and assign storage capacity boundaries to **20 GB** or **30 GB**.
4. Expand the configuration sub-properties for the virtual peripheral array line item labeled **CD/DVD Drive 1**. Adjust its source mapping mechanism from *Host Device* directly over to **Datastore ISO file**. Select your uploaded server binary image file from the datastore browser interface.
5. **Crucial Validation Step:** Verify that the activation checkbox parameter configuration titled **Connect at power on** is explicitly toggled to true (**[X]**). This signals the host environment to load the virtual install media immediately upon power delivery.
6. Conclude setup by clicking **Finish**. Proceed to launch the newly minted instance configuration framework container shell by toggling the green **Power On** execution switch. Click inside the visual workspace framework container module viewport to link keyboard polling controls.

Phase 4: Chronological Ubuntu Installer Reference

Since the mouse interface is unavailable during text-based environment setup, navigations require targeted keyboard tracking patterns. Move between selection nodes with the **Arrow keys**, cycle check parameters with the **Spacebar**, and validate operations via the **Enter key**.

Installation Phase Screen	Action / Technical Choice Checked	System Mechanics Alteration
Installation Base Type	Select and verify option (X) Ubuntu Server . Avoid the minimized alternative layout.	Instructs the operational deploy script engine to unpack the foundational management toolchains and system drivers needed for normal network administration.
Network Mapping Setup	Retain structural parameters defined automatically by device controller interface tracker ens34 . Bypass the secondary module step labeled <i>[Create bond]</i> .	Signals the virtual interface module to pool local node access paths and register an independent IPv4 lease (e.g., <code>192.168.50.35</code>) directly from your router pool.
Proxy Address Map	Keep string field completely Blank . Navigate downstream past options.	Confirms that the host node has direct layer-3 structural egress routing access paths straight to your network gateway device without needing a proxy server.
Ubuntu Archive Mirror	Retain geographical package archive URL route configurations. Ensure validation diagnostics output reads "passed tests".	Tests link throughput and confirms access paths to core repository structures, enabling your system to seamlessly retrieve critical packages and dependencies.
Guided Storage Layout	Retain defaults (X) Use an entire disk and check option [X] Set up this disk as an LVM group . Proceed past the destructive alert block verification check.	Directs the installer layout manager to format the empty virtual volume block array and map it inside a Logical Volume Manager (LVM) framework for clean storage scaling later.
SSH Configuration	Explicitly change the inactive checkbox option parameters to select [X] Install OpenSSH server . Keep authorization type targeting passwords.	Deploys and initializes the remote system shell daemon. This allows secure network access control straight to the Linux terminal using external tools (like PuTTY or Windows Terminal) over the network.
Featured Server Snaps	Leave all optional software container frameworks completely blank [] . Scroll past the index matrix layout structure blocks.	Skips bundling bloated auxiliary application architectures at boot time, ensuring the newly created virtual template machine environment

Installation Phase Screen	Action / Technical Choice Checked	System Mechanics Alteration
		remains completely clean and performance-optimized.

NEXT STEP PROTOCOL: DEPLOYMENT CONCLUSION

Upon committing the configurations on the final Server Snaps panel screen, the underlying automation engine partitions the assigned disk sectors and clones operating system assets. Watch the runtime text output frame. Once structural replication processes terminate, a final administrative button labeled [**Reboot Now**] will appear. Activating this option safely finishes installation, unmounts your virtual installation disk image media map, and boots up your new production server console!